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Educational Reviews

Reprint of: Acute Graft-versus-Host Disease: Novel Biological Insights[☆]



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ABSTRACT

Graft-versus-host disease (GVHD) continues to be a leading cause of morbidity and mortality after allogeneic hematopoietic stem cell transplantation. Recent insights into intestinal homeostasis and uncovering of new pathways and targets have greatly reconciled our understanding of GVHD pathophysiology and will reshape contemporary GVHD prophylaxis and treatment. Gastrointestinal (GI) GVHD is the major cause of mortality. Emerging data indicate that intestinal stem cells (ISCs) and their niche Paneth cells are targeted, resulting in dysregulation of the intestinal homeostasis and microbial ecology. The microbiota and their metabolites shape the immune system and intestinal homeostasis, and they may alter host susceptibility to GVHD. Protection of the ISC niche system and modification of the intestinal microbiota and metabolome to restore intestinal homeostasis may, thus, represent a novel approach to modulate GVHD and infection. Damage to the intestine plays a central role in amplifying systemic GVHD by propagating a proinflammatory cytokine milieu. Molecular targeting to inhibit kinase signaling may be a promising approach to treat GVHD, ideally via targeting the redundant effect of multiple cytokines on immune cells and enterocytes. In this review, we discuss insights on the biology of GI GVHD, interaction of microflora and metabolome with the hosts, identification of potential new target organs, and identification and targeting of novel T cell–signaling pathways. Better understanding of GVHD biology will, thus, pave a way to develop novel treatment strategies with great clinical benefits.

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INTRODUCTION

Acute graft-versus-host disease (GVHD) is an immunologically mediated process, involving donor T cell responses to host alloantigens and the dysregulation of inflammatory cytokine cascade after allogeneic hematopoietic stem cell transplantation (allo-SCT). Studies on GVHD pathophysiology have been primarily focusing on mechanisms of T cell activation in the induction phase of GVHD. Recent studies on the fundamental mechanisms of the target injury highlight complex interactions between different effector cell type and target cells in peripheral tissue. Particularly, gastrointestinal (GI) tract damage and subsequent alteration of the intestinal homeostasis play important roles in amplifying GVHD [1].

Emerging data suggest that alterations in the intestinal microbiota and metabolome also play crucial roles in modulating the severity of acute GVHD.

Current GVHD prophylaxis and treatment are only partially effective, with an increased risk for infections, disease relapse, and long-term adverse effects. Data suggest that use of T cell–directed immunosuppressants potentially inhibits tolerance induction, at least in part by suppressing regulatory T cell (Treg) homeostasis [2]. Cytokines and chemokines are the major players in GVHD and blockade of TNF- α , IL-6, and C-C chemokine receptor type 5 (CCR5) has been tested based on data from mouse models [3–7]. Recent progress in our understanding of signaling pathway and molecular targeting enables us to target the redundant effect of multiple cytokines. Thus, better understanding and a more targeted approach of signaling pathways in T cells with a newer class of immunomodulatory approaches could lead to more effective control of GVHD. In this review, we discuss newer experimental insights on the biology of GI-GVHD and the interaction of intestinal microbiota and metabolome with the host and novel strategies for T cell–signaling manipulation in the regulation of GVHD.

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EMERGING CONCEPTS ON TISSUE INJURY IN GVHD

Disruption of the Intestinal Homeostasis and Microbial Ecology in GI GVHD

Intestinal GVHD is characterized by severe villous atrophy and crypt degeneration, but the crypt cell degeneration has been suggested to be the initial lesions of intestinal GVHD [8–10]. Intestinal stem cells (ISCs) and their niche Paneth cells reside within the crypts and play a pivotal role in both physiologic tissue renewal and regeneration of the injured epithelium. We and others have tested the hypothesis that ISCs and their niche could be primary targets in GVHD in mice [11–13]. Irradiation induces apoptosis of ISCs within 24 hours while sparing Paneth cells, but ISCs rapidly recover and restore the normal architecture of the small intestine in the absence of GVHD [12,14]. However, once GVHD develops, both ISCs and Paneth cells do not recover, indicating that the ISC-niche system is the target in intestinal GVHD (Figure 1) [11–13].

Secretory cells, such as Paneth cells and goblet cells, play a critical role in maintaining intestinal microbial ecology and protecting hosts from pathogens. Paneth cells shape the microbial composition by secreting a range of antimicrobial peptides, such as α -defensins [15]. In GVHD, Paneth cell loss results in a reduced secretion of α -defensins, leading to intestinal dysbiosis [13,16–19]. Goblet cells shield the epithelium from luminal bacteria by secreting mucin [20]. Goblet cell injury in GVHD permits translocation and dissemination of dominant luminal pathogens, which further accelerates GVHD and creates a vicious cycle of GVHD and infections [21,22]. These mechanisms explain why GI GVHD is often prolonged and refractory to treatment and represents a risk factor for septicemia. These findings have been confirmed in human studies; loss of Paneth cells and dysbiosis are hallmarks of GI GVHD and associated with increased

nonrelapse mortality [17,23–26]. The effects of innate cells, such as neutrophils, monocytes, and innate lymphoid cells, in target injury are also now becoming known [11,27,28].

The intestinal microbiota has been given a putative role in the pathogenesis of GVHD since the pioneering studies in the early 1970s [29,30]. Although the focus so far has been on bacteria and their metabolites, several clinical observations have also suggested a role for *Candida* in the pathogenesis of GVHD; *Candida* colonization in the gut is a risk of acute GVHD (aGVHD) and antifungal prophylaxis with fluconazole reduced aGVHD [31,32]. However, a mechanistic link between fungal infection and GVHD has never been demonstrated. A component of the *Candida* cell wall, α -mannan, is recognized by the C-type lectin receptor Dectin-2 and stimulates macrophages to produce cytokines, such as IL-6 and IL-23, which drive a Th17-skewed response [33]. A lung-specific chemokine/cytokine environment in GVHD leads to accumulation of Th17 cells in the lung [33,34]. α -Mannan or heat-killed *Candida albicans* exacerbate systemic aGVHD, particularly in the lung [33]. Thus, not only bacteria but also fungi residing on the mucosal surface play an important role in GVHD. Th1 and Th17 responses may differentially contribute to tissue-specific aGVHD. However, studies using an IL-17 fate-mapping reporter highlight the complex plasticity and heterogeneity in cytokine expression in effector T cell subsets involved in the development of GVHD [33–36], which will need further elucidation to guide rational development of preventive and therapeutic interventions.

Protection of the ISC Niche System as a Novel Approach to Restore Intestinal Homeostasis and Control GVHD

Attention has now begun to focus on protection of the ISC niche system to regulate GVHD. ISCs express IL-22 receptors

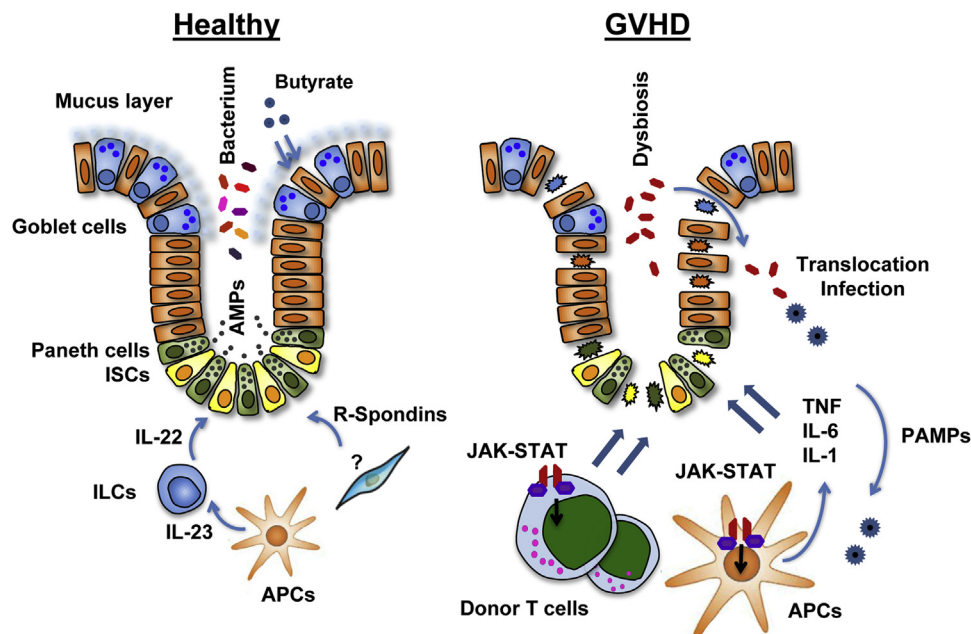


Figure 1. Homeostasis of self-renewing small intestinal epithelium results from ISCs, which locate at the base of the crypts and interspersed between their niche Paneth cells. IL-22 produced from innate lymphoid cells (ILCs) upon stimulation of IL-23 from antigen-presenting cells (APCs) and R-spondins play a role in self-renewal and differentiation of ISCs. Paneth cells maintain intestinal microbial ecology by secreting antimicrobial peptides (AMPs) into the lumen. Goblet cells shield the epithelium from luminal bacteria by secreting mucin. Intestinal homeostasis depends on proper interaction between the mucosal immune system and intestinal microbiota and its metabolites, such as butyrate. After allo-SCT, donor T cells and inflammatory cytokines, such as TNF, IL-6, and IL-1, damage the ISC-niche system. Paneth cell loss results in a reduced secretion of AMPs, leading to intestinal dysbiosis. Tissue injury permits translocation and dissemination of dominant luminal pathogens and pathogen-associated molecular patterns, which further accelerates GVHD by propagating a proinflammatory cytokine milieu. JAK-STAT pathways are required for responses to multiple cytokines and, thus, represent potential new targets in GVHD.

[11]. Recipient IL-22 deficiency increases crypt apoptosis, loss of ISCs, and GVHD mortality [11]. Administration of IL-22 may be a promising strategy to protect ISCs [37]. Wnt agonist R-spondin1 (R-Spo1) [38] is an essential growth factor for ISCs [39]. Brief administration of R-Spo1 interrupts acceleration of GVHD by protecting ISCs from conditioning, facilitating wound healing, and inhibiting the inflammatory cytokine milieu in mice [12]. We also showed that R-Spo1 stimulates ISCs to differentiate into secretory cells. R-Spo1 thus inhibits the GVHD-associated dysbiosis and translocation of the luminal bacteria by promoting Paneth cell secretion of α -defensins and preserving the GI mucus layer. Interestingly, the survival benefit mediated by R-Spo1 was in part abrogated when mice were treated with oral broad-spectrum antibiotics, suggesting a link between GVHD and intestinal commensals. Intestinal homeostasis depends on proper interaction between the mucosal immune system and intestinal microbiota and its metabolites. Identification of the specific bacteria and associated products that affect transplantation outcome is now an active area of investigation [40]. Such a novel pharmacological approach to protect the ISC niche system and modulating the intestinal environment may shape host immune system toward less GVHD while preserving graft-versus-leukemia (GVL) activity [41].

Previously Unrecognized Target Organs of Acute GVHD

Although aGVHD typically targets the skin, GI tract, and liver, recent experimental studies suggest previously unrecognized organs as potential targets of aGVHD. It has been shown that GVHD is a risk for neurological complications and infertility [42,43]. Recent studies in mice and primates demonstrated significant donor T cell infiltration of the central nervous system with damage of neurons and neuroglia during aGVHD, leading to behavioral abnormalities [44,45]. We also found donor T cell infiltration in the ovary with apoptosis of ovarian cells, leading to impairment of ovary functions and infertility. Pharmacological GVHD prophylaxis preserved fertility after allo-SCT with a minimally toxic conditioning regimen [46]. These results may have important clinical implications in young female transplant recipients with nonmalignant diseases, where minimally toxic conditioning regimens are used. Such increasingly appreciated manifestations of GVHD in animals deserve further clinical scrutiny.

MICROBIAL METABOLITES AND HISTONE ACETHYLATION OF GI EPITHELIUM IN GVHD

Histone Deacetylase Inhibitors and GVHD

Histone acetylation regulates transcriptional activation. Lysine residues at the amino-terminus of histone H3 and H4 tails are acetylated by histone acetyltransferase enzymes or deacetylated by histone deacetylase (HDAC), leading to modifications in DNA accessibility. HDAC inhibition results in the accumulation of hyperacetylated histones, thereby altering the patterns of gene expression [47]. Beside their role in cancer, HDAC inhibitors (HDACi), at lower and non-cytotoxic concentrations, possess anti-inflammatory and immunoregulatory effects [48]. Animal studies of allo-SCT have shown that HDACi suppress proinflammatory cytokine production, reduce GVHD, and preserve GVL by modulating indoleamine-2,3-dioxygenase-dependent innate immune and allo-stimulating functions of antigen-presenting cells in a STAT-3-dependent manner [47,49]. HDACi also enhance natural Treg functions [47,50]. A recent single-arm, phase I/II, first-in-human clinical trial with

HDACi vorinostat in adult patients with high-risk hematological malignancies after HLA-matched related donor SCT showed potential for preventing severe GVHD [51]. Histone acetylation regulates the epigenetic landscape of GVHD target tissues as well [52]. However, the role of histone acetylation and its dependent epigenetic regulation in GVHD target tissues, specifically, the GI tract, and the impact on overall severity of GVHD, remains largely unknown.

Intestinal Metabolome in GVHD

The microbiota are known to perform key metabolic functions; they metabolize not only material directly ingested by the host but also produce byproducts of its own metabolism [53]. The intestinal metabolome thus consists of products from discrete host metabolism, microbial metabolism, and mammalian-microbial cometabolism [54]. The impact of microbiota-derived metabolites is being increasingly appreciated, specifically in intestinal homeostasis [55]. Recent studies show that colonocytes from germ-free mice are in an energy-deprived state, which in turn result in autophagy [56].

In addition to the effects of microbial metabolites on nonimmune cells, the impact of the metabolome on immune cells is now becoming increasingly understood. Recent studies have shown that 17 rationally selected strains of *Clostridia*, known to produce the short chain fatty acid (SCFA) butyrate, directly result in the increased presence of Tregs in the gut [57]. Tregs play a critical role in maintaining GI homeostasis by modulating inflammatory responses via the release of the anti-inflammatory molecule IL-10, which also directly impact macrophages [58]. Diet plays a role in regulating rapid changes in the taxonomic composition of the gut microbiota and an appropriate diet conducive to homeostatic microbiota may be an important factor when treating comorbidities, as the associated metabolome is correspondingly altered [59]. Butyrate, a critical SCFA, is an HDACi. Together, these data suggest the microbial metabolome, more specifically intestinal butyrate, might impact GVHD, although this hypothesis has heretofore not been formally tested.

Many recent studies have made strides in identifying the taxonomic composition and activity of the host microbiome. Although it is increasingly evident that alterations in the microbiome correlate with many disease states, including GVHD [13,17,19], the mechanism through which these alterations confer their effects is poorly understood. The most studied microbial metabolites SCFAs are produced by microbial fermentation of complex polysaccharides in the gut and are subsequently absorbed by the intestinal epithelium [60]. Butyrate is utilized as a major energy source for intestinal epithelial cells [60]. Thus, microbial metabolites may play an important role in maintaining the health of the physical barrier of the gut epithelium as well as protecting the host from the influences of pathogen-associated molecular patterns and damage-associated molecular patterns. SCFAs may have a particularly important role in the protection of the host from GVHD. Observations from our lab, exploring the generation of SCFAs and the impact on intestinal tissues and GVHD, will be discussed [61].

Briefly, inflammation after allo-SCT reduced histone acetylation in the IECs of the hosts in association with reduced butyrate within the IECs. We explored whether the alterations in acetylation and butyrate levels within the IECs after allo-SCT correlated with luminal SCFA levels that are exclusively made by intestinal microflora after allo-SCT.

Surprisingly, the concentrations of the SCFAs were not significantly different in serum or tissues, such as the spleen and liver, of allogeneic animals compared with syngeneic animals and naïve controls. However, the greatest and the only statistically significant difference was, unexpectedly, observed in just 1 SCFA, butyrate, which was significantly decreased only in the intestinal tissue after allo-SCT. The reduced butyrate in CD326⁺ IECs after allo-SCT contributed to decreased histone acetylation, because this was restored upon local administration of exogenous butyrate. Furthermore, increasing local delivery of butyrate resulted in improved IEC junctional integrity, increased antiapoptotic proteins, decreased GVHD, and improved survival. In addition, alteration of the indigenous microbiota with 17 rationally selected strains of high butyrate-producing *Clostridia* [57] also decreased GVHD and increased survival after allo-SCT. These data demonstrate a heretofore unrecognized role of microbial metabolites and suggest that local and specific alterations of microbial metabolites have direct salutary effects on GVHD target tissues and can mitigate its severity.

KINASE INHIBITION FOR GVHD TREATMENT

Identification and Targeting of Novel T Cell signaling Pathways

Besides the role of kinases in cancer, abnormal activation of certain kinases is a hallmark of many inflammatory diseases. Examples are abnormal PI3 kinase delta activation in activated PI3 kinase delta syndrome, increased Janus kinase (JAK) activity in rheumatoid arthritis and ulcerative colitis, or abnormal spleen tyrosine kinase (SYK) activity in immune thrombocytopenic purpura and rheumatoid arthritis. Consistent with a role of many kinases in different immune cells during autoimmunity, a number of kinases have been reported to be targets in GVHD. Analyses of the role of different kinases in GVHD were done via the use of kinase inhibitors, mostly in a prophylactic setting in different murine models of GVHD or by the use of gene-targeted mice lacking the respective kinase. Kinases identified to be potential targets in GVHD by these approaches include rho-associated coiled coil kinase [62], protein kinase C (PKC) α [63], PKC θ [63,64], SYK [65–68], JAK 1/2 [69–72], JAK3 [73], cyclin dependent kinases [74], mitogen-activated protein kinase kinase [75], and Bruton's tyrosine kinase [76]. Strong evidence for a preclinical and clinical efficacy exist for SYK and JAK1/2 inhibition in GVHD, based on multiple publications of independent groups [65–72]. SYK is an intracellular nonreceptor tyrosine kinase involved in multiple signaling events of immune cells and has shown activity in aGVHD [65] and chronic GVHD (cGVHD) [66,67]. Intact SYK is required for extracellular signal-regulated kinases (ERK) activation and nuclear factor of activated T cells (NFAT) transcriptional activation [77], and recognition of certain peptides affords SYK for TCR signaling [78]. In our previous work, we showed that SYK is required for the development of aGVHD by using a pharmacological inhibitor [65], which was later extended by other groups to cGVHD [66–68]. SYK inhibition with fostamatinib reduced in vivo expansion of donor T cells and dendritic cell (DC) migration in mice developing aGVHD without blocking antiviral and GVL responses [65]. In cGVHD, fostamatinib treatment normalized germinal center formation and decreased activated CD80/86⁺ DCs [67]. The effects of SYK inhibition could be connected to the hematopoietic system, as bone marrow-specific SYK deletion reduced established cGVHD [67]. Beside their investigation

in mice, Flynn et al. also showed that SYK inhibition was effective at inducing apoptosis in human B cells derived from individuals with cGVHD [67].

JAK Inhibition for GVHD Treatment

A group of kinases that were recently shown to be potential targets in GVHD are JAK 1, 2, and 3 [69–72]. When these kinases are activated, signal proteins of the STAT family are phosphorylated, which act as transcription factors for target genes that serve proinflammatory roles. Among other functions, JAK1 and 2 are required for responses to multiple cytokines [79]. Signaling through the common γ -chain, a constituent of the receptor complexes for 6 different interleukins, including IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21 [80], is mediated via JAK1. Signaling through the common γ -chain was recently identified as a therapeutic target in aGVHD and cGVHD [72]. Blockade of multiple cytokines by inhibition of the downstream signaling could avoid the redundant effect of most cytokines. This could help overcome the unsatisfactory results reported when a single cytokine is inhibited. Another event where JAK1/2 inhibition may interfere with GVHD is DC activation, which was shown to depend on JAK1/2 activation [81]. Additionally, JAK1/2 signaling plays a central role in the activation of neutrophils [82], which we and others connected to the pathogenesis of aGVHD [83–85]. Based on the preclinical evidence, the efficacy of the JAK1/2 inhibitor ruxolitinib was recently assessed in a retrospective survey [86]. Ninety-five mostly heavily pretreated patients who had received ruxolitinib as salvage therapy for steroid refractory (SR) aGVHD or cGVHD were reported [86]. The 6-month-survival of the patients treated with ruxolitinib was 79% (95% confidence interval, 67.3% to 90.7%) and 97.4% (95% confidence interval, 92.3% to 100) for SR aGVHD and SR cGVHD, respectively. Trials of SR GVHD are heterogeneous, making direct comparison of different agents difficult [87]. The weighted average 6-month survival estimate across 25 retrospective or phase II trials on SR aGVHD was 49% [87]. The largest study analyzed the effect of antithymocyte globulin as a second-line treatment and included 79 patients [88]. The 6-month survival estimate for this study was 44% [88]. The use of ruxolitinib for SR GVHD was connected to an overall response rate of 81.5% (44 of 54) in SR aGVHD, including 25 complete remissions (46.3%), whereas for SR cGVHD, the overall response rate was 85.4% (35 of 41) [86]. When only patients responding to ruxolitinib were analyzed, the rate of GVHD relapse was 6.8% (3 of 44) and 5.7% (2 of 35) for SR aGVHD and SR cGVHD, respectively [86]. Major side effects during ruxolitinib treatment included cytopenia and cytomegalovirus reactivation in both SR aGVHD and SR cGVHD patients, which were, however, comparable to other reported immunosuppressive approaches, including steroids, infliximab, alemtuzumab, mycophenolate mofetil, or cyclosporine [89–91].

A major concern of pharmacological immune modulation via kinase inhibition is the loss of GVL activity. Reduced GVL effects were observed, for example, when PKC α and PKC θ proteins were inhibited [62]. Relapse of the underlying malignancy in ruxolitinib-treated patients was low, as only 9.3% and 2.4% of the patients with aGVHD or cGVHD, respectively, had disease recurrence [86]. Major limitations for a broad clinical application of kinase inhibitors in allo-SCT are the moderate specificity of many compounds as off-target inhibition can lead to considerable side effects. Additionally, GVL

and antimicrobial effects need to be preserved, as purely immunosuppressive effects would not lead to an advantage over already established immunosuppressive drugs that are routinely used in clinical practice.

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